Kevin Pulikkottil

San Jose, CA | 669-253-8966 | kevinpul101@gmail.com
linkedin.com/in/kpulik | github.com/kpulik | kpulik.github.io/Portfolio

EDUCATION

University of Illinois Urbana-Champaign — Urbana-Champaign, IL
Master of Computer Science — *Starting Spring 2027*

The University of Texas at Dallas — Richardson, TX
Bachelor of Science in Computer Science — *May 2026*

Relevant Coursework: Computer Networks & Lab, Operating Systems, UNIX Systems Programming, Database Systems, Data Structures & Algorithms, Software Engineering

RELEVANT EXPERIENCE

Toyota Secure Vehicle Communications | UTDesign Capstone — Jan 2026 - May 2026
Vehicle Cybersecurity Engineer | Toyota Tsusho Systems-sponsored capstone team

- Engineered an X448 Elliptic Curve Diffie-Hellman key exchange in an OP-TEE Trusted Application on Arm64 QEMU, enabling rekey-capable EV-to-charger sessions for an ISO 15118-20 prototype.
- Stabilized X.509 certificate generation and validation by removing a custom CMAC extension across the builder, parser, verifier, and Trusted Application.
- Resolved Arm64 build failures across four source files and helped automate Trusted Application rebuilds and testing under QEMU.
- Developed a browser-based dashboard demonstrating live man-in-the-middle attack detection with a Raspberry Pi attacker node; presented the system to corporate stakeholders at UTDesign EXPO.

PROJECT EXPERIENCE

AutoSIEM | Detection Engineering and SOC Investigation — Python, SQLite

- Developed an open-source SIEM MVP with MITRE ATT&CK-mapped rules, seven UEBA signals, entity risk scoring, and 24-hour incident correlation across users, hosts, and IPs.
- Built log ingestion for Syslog/CEF, Okta, Entra ID, GitHub, and AWS CloudTrail; added role-based access controls, auditable incident triage, and approval-gated AI response proposals.
- Integrated Sigma detection content; repository validation imported 895 community rules, and the project roadmap reports 1,174 passing automated tests.

Security Toolkit API | Security Assessment Service — Python, FastAPI

- Developed and deployed a FastAPI service on Render and RapidAPI for password breach checks via HIBP k-anonymity, SSL/TLS analysis, passive TLS scanning, and GitHub secret detection.
- Implemented API rate limits, authorization attestation for scans, masked secret findings, and structured error responses.

Secure Home Lab | Network Security — Dec 2025 - Present

- Deployed Linux-based Pi-hole DNS filtering and Tailscale access, reporting approximately 99% less ad traffic; configured firewall rules and inspected DNS query logs.

MemPalace | Merged Open-Source Contribution — Apr - Oct 2026

- Contributed merged PR #2694 (Oct 2026): implemented opt-in EmbeddingGemma 2 multimodal search across text, code, images, audio, and video, with Apple MPS inference and CPU fallback.
- Added media ingestion, embedding-space identity checks, and safe Chroma index recovery; all 13 reported cross-platform CI checks passed.
- Earlier merged PR #315: fixed MCP CLI setup guidance and added Pytest regression tests for custom palace paths.

CERTIFICATIONS & LEADERSHIP

CompTIA Security+ | Earned April 2026 (SY0-701)
Eagle Scout | Led a volunteer community build, raised approximately $900, and coordinated construction of three planter boxes.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, Bash
Security & Networking: SIEM, UEBA, Sigma rules, MITRE ATT&CK, TCP/IP, DNS, TLS, firewalls, X.509, X448, OP-TEE
Systems & Tools: Linux, Windows, Git, Raspberry Pi, Tailscale, Pi-hole, Arm64 QEMU, Pytest, FastAPI, AWS IAM fundamentals